Abstract

Algorithmic trading (a.k.a. high-frequency trading) is generally defined as using computer-generated algorithms to create and execute orders on marketplaces. Recently, such algo-trading strategies are increasingly being associated with the negative impact on capital markets – both from a technological as well as a business perspective. However, there are positive effects of algorithmic trading too – such as increased liquidity and the elimination of market inefficiencies – that far outweigh the potential negative effects. Regardless of this, algo-trading has seen increasing popularity and acceptance on most of the major global markets, demanding well thought-out strategies that actually help traders and investors make more than a reasonable return on their investments.

Objective of this paper is to:

a) Categorize the types of algorithmic trading and describe their implications for capital markets.
b) Describe how to accommodate algorithmic trading in markets, while minimizing potential adverse effects. Discuss technology options and architectures to realize the same.
c) Describe the use of special-purpose orders in the context of algorithmic trading.

Keywords: Algorithmic trading, High-frequency trading, Automated trading, Flash trading

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INTRODUCTION

Electronic (capital) markets have become more pervasive and are characterized by a relentless drive towards faster decision making and execution. In view of this, both the developed and emerging markets have seen significant increase in trading volumes through complex algorithms executed on massive computers – more popularly known as 'algorithmic trading' or algo-trading. About seven billion shares exchange hands every day on US equity markets, of which around two-thirds are traded by computer generated algorithms based on quantitative models that crunch mountains of data to predict gains while trying to minimize or reduce risk. Further, almost half of Asia Pacific markets are traded electronically while the emerging markets are slowly catching-up to this trend.

Recently, such algo-trading strategies are increasingly being associated with the negative impact on capital markets – both from a technological as well as a business perspective. However, there are positive effects of algorithmic trading too – such as increased liquidity and the elimination of market inefficiencies – far outweigh the potential negative effects.

Despite the bad press, it's a fact that high-frequency trading has seen increasing popularity and acceptance on most of the major global markets, demanding well thought-out strategies that actually help traders and investors make more than a reasonable return on their investments.

OBJECTIVE(S)

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ORGANIZATION OF SECTIONS

This paper is organized as follows:

1) Problem formulation
2) Description
3) Assumptions
4) Related research
5) Authors' own contributions
   - Algo-trading categories, its implication on markets and their consequences
   - Accommodating algo-trading categories
   - Technology framework options
   - Conceptual architecture framework
   - Conceptual architecture model
6) Conclusion
7) References

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1. PROBLEM FORMULATION

Algorithmic trading is becoming a global phenomenon in most mature capital markets, but the subject is a complex one to comprehend, as there are major differences in market maturity in different parts of the world. It's imperative to be cognizant of the fierce competition from brokers, electronic execution specialist firms who continue to invest heavily in algo-trading platforms, thereby making it increasingly difficult for others to compete. Most of the current research on Algo-trading is around market quality, latency, market making, elimination of market inefficiencies, implications on liquidity, cost etc., leaving not much ground being covered in its overall implications with regard to the broader capital markets.

2. DESCRIPTION

2.1.1 What is Algorithmic (or) High-frequency trading?

It is the high-speed execution of automated trading strategies, in which large number of trades are conducted in short time spans in order to profit from pricing or other market inefficiencies. In simple terms, it is the use of programs and computers to generate and execute (large) orders in markets with electronic access. Securities and Exchange Board of India (SEBI) has succinctly defined Algorithmic-trading as "any order that is generated using automated execution logic."

Capital markets are global and inter-connected but vary in size, maturity and technological capabilities. Initially in the early '90s, algorithms were used as proprietary tools by institutional investors in the U.S. to create new markets aside from NYSE and NASDAQ which, has since led to the explosion of high-frequency trading globally. This has resulted in orders pouring from institutional investors, hedge funds and several Wall Street trading desks. In addition to the primary objective of maximizing profits, algo-trading helps control execution costs and limit market risks.

2.1.2 Issues faced in Algo-trading

Institutional clients usually trade large volumes in equities, forex, fixed-income, commodities, derivatives and emerging market asset classes. These volumes (quantity) and values (trading currency) are often larger than what the market can absorb without impacting the price. Therefore, there is a compelling need for such large orders to be split or sliced into smaller orders to be executed electronically over the course of seconds (even milliseconds!), minutes, and hours in a day, which lends itself as a challenge from both technology selection and architectural (system) design and implementation perspective.

2.1.2.1 Most commonly observed issues are –

a) The decision of how to split or slice an order into smaller (tradeable/executable) chunks.
b) Order execution strategies and at what prices?
c) Choice of algorithms – alpha preserving or alpha creating?
d) Choice of technology and architectural components.
e) Achieving the 'best possible price' for each of those tradeable/executable orders.
f) Creation of dark pools

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2.1.2.2 Several automated trading worlds

There are a number of automated trading venues (markets) and market participants face innumerable challenges that are widely varied and deeply impacted by a range of factors – both internal and external to the exchange. Some of the major trading venues for US Equities are set forth below –

g) ARCA-NYSE (ex-Archipelago) – electronic trading platform of NYSE
h) Better Alternative Trading System, commonly known as BATS
i) Direct Edge
j) ISE – International Securities Exchange

3. ASSUMPTIONS

The following assumptions form the basis of this paper –

a) Technological innovation allows for trading strategies (orders) to be executed rapidly leading to the growth of algo-trading in every market in which it exists.
b) Reduced barriers to entry have increased access for new market participants.
c) The proliferation of new trading venues (exchanges) creates arbitrage opportunities for algo-traders.
d) Regulatory changes have led to the introduction of new trading rules (eg: decimalization), which has led to certain price discrepancies.

4. RELATED RESEARCH

Over the past few years, a good amount of research has been carried out in this field and increasingly, with the advent of technologically-driven globally inter-connected markets there will be even greater research undertaken in this area resulting in hopefully a better understanding of –

- Algorithmic trading strategies used by high-frequency traders.
- Its implications with respect to global capital markets.
- How various quantitative models behave/interact to arrive at an optimal order execution strategy?
- Profitability, and their relationship with the characteristics of the overall market such as:
    - Liquidity
    - Price discovery
    - Volatility

In a 2010 research carried out by **Kellogg School of Management, Northwestern University, Jonathan A.Brogaard** analyzed a unique dataset to study the strategies used by twenty six High-frequency trading (HFT) firms that led to the following observations with respect to its implication in market quality –

1. HFTs tend to follow a price reversal strategy driven by order imbalances.
2. HFTs earn gross trading profits of approximately $2.8 billion annually and their strategies are more correlated with each other than are non-HFTs.
3. HFTs' trading levels change only moderately as volatility increases. That said, they may also dampen intra-day volatility.

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4. HFTs add substantially to the price discovery process and provide the best bid and offer quotes for a significant portion of the trading day.

It's important to highlight that, while the aforementioned research examines several key questions regarding HFT, it does not necessarily attempt to analyze certain other pertinent aspects such as flash quotes, latency arbitrage, quote stuffing or the order book dynamics of HFTs.

Another recent research (February 2013) conducted by **Ciamac C.Moallemi** from **Graduate School of Business, Columbia University** and **Mehmet Sağlam** from **Bendheim Center for Finance, Princeton University** described a theoretical model for quantifying latency – the delay between a trading decision and the resulting trade execution – which is considered a major challenge for both HFT firms and algo-traders.

Their research has profound implications in the context of capital markets since over the past few years, competition between exchanges through significant technological investments have led to dramatic improvements in latency – a key differentiator when it comes to algo-traders choosing one exchange over the other to execute trades with the lowest possible latency (a.k.a. low-latency trade execution).

5. AUTHORS' OWN CONTRIBUTIONS

Most of the research currently being pursued in Algo-trading is around market quality, latency, market making, elimination of market inefficiencies, implications on liquidity, cost etc. To the best of my knowledge, not much ground has been covered in its overall implications with regard to capital markets. In order to understand the overall implications of algo-trading with respect to capital markets, it's important that the paper addresses the objectives outlined in Page 2 of this publication.

5.1 Algo-trading categories

Algo-traders incorporate a wide variety of trading strategies on a daily basis, some of which are widely known – e.g. arbitrage between index futures and underlying instruments forming part of the index. Others, being proprietary in nature are closely guarded secrets at Hedge funds and Algo-trading desks of investment banks. The aim of this paper is not to present an exhaustive list of trading strategies commonly used by high-frequency traders, but to categorize them logically into –

a) Alpha – preserving strategies [αP]

In **αP strategies**, algorithms are used when a trader or investor has – through fundamental analysis, research, intuition or in some other manner – reached a trading decision and is about to execute it electronically. This could even be as simple as breaking up a large order into smaller order chunks (split or slice) executed over a certain period of time.

These algorithms deal with reducing market impact, thus reducing slippage (or failure) when executing a particular chosen strategy. These algorithms are usually considered simpler than alpha-creating [αC] ones.

b) Alpha – creating strategies [αC]

In **αC strategies**, the algorithm includes a diverse set of algorithms that by themselves try to create alpha [α], i.e. it takes the volatility (price risk) of a portfolio or a fund and quickly compares its risk-adjusted performance to a benchmark index for eg: DJIA, FTSE 100, S&P 500, NASDAQ Composite

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etc., and then computes the excess returns of the portfolio or fund relative to the return of the benchmark index – which is the alpha [α].

While αC algorithms do provide additional liquidity in the marketplace, they could result in some negative consequences – both from a technical and business perspective – depending on the market model being used and the technical solution being deployed to execute a particular trading strategy.

5.1.1 αP algorithms and its implications on markets

Most αP algorithms involve spreading out an execution of a large order both over time and execution venues i.e. exchanges, with the primary goal being to minimize slippage when executing the order. A slippage occurs due to two main reasons:

- Lack of liquidity in the marketplace.
- Impact on the market – wherein the market is believed to be shifting as a result of new information created by the order that is being executed.

From a technology perspective, spreading out the execution of an order over time and execution venues involves chopping the bulk order into small pieces (splitting or slicing) to be then fed into various execution venues (exchanges) gradually over a period of time.

5.1.2 Consequences

- **Increased transaction flow** as a result of splitting or slicing order into smaller chunks, which in effect could be negligible but is prevalent.
- **Decreased liquidity** as a result of withholding orders instead of entering them directly into the market, which could be quite significant in the overall context of capital markets.

5.1.3 Accommodating αP algo-trading

This paper deals with the following aspects with regard to accommodating alpha-preserving [αP] algorithmic trading:

a) **Increased transaction flow** – Apparently, increased order flow resulting from splitting or slicing orders is usually not large enough to pose a significant performance issue in markets. However, this could result in increased amount of trades that could further render billing methods based on '*charging per trade*' somewhat complicated. A rather simplistic way to mitigate this is to '*nett the trades*', which drastically reduces the number of trades that the market participants have to deal with the back-office systems and applications.

b) **Decreased liquidity** – One of the pertinent causes of slippage due to market impact is the lack of transparency. A few techniques for order entry that are widely prevalent are iceberg orders (quantity publically shown is randomized every time it is exhausted and refilled from the hidden quantity), dark orders (orders that are active and match against the visible book but are not shown themselves) and delayed publication of larger trades.

5.1.4 αC algorithms and its implications on markets

Although αC algorithms undeniably provide a lot of liquidity into the markets, if looked upon in seclusion, they may however as a consequence, have negative effects on the marketplace as a whole.

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5.1.5 Consequences

- **Increased transaction flow** (*either* by design *or* erroneously) – it is imperative to note that some αC algorithms produce large transaction flows, often with a high order-to-trade ratio, which implies that the marketplace needs to scale up to be able to handle the increased transaction flow.
- **Could discourage market making** – through αC algorithms that could take advantage of weaknesses and inefficiencies in the market making applications.
- **Could discourage institutional investors** – because some αC algorithms take unfair advantage of the information leakage that is inherently present, when executing large or bulk orders.

5.1.6 Accommodating αC algo-trading

This paper attempts to deal with the following aspects with regard to accommodating alpha-creating [αC] algorithmic trading:

a) **Increased transaction load** – One way of controlling transaction flow is to throttle in-bound transaction flows, as depicted in *Figure 1* below. A market participant is assigned a maximum transaction rate for eg: '*1000 txns/sec*'. Now, if the rate is exceeded, then transactions from that market participant are either delayed or rejected. A widely practiced strategy by participants is to have the possibility to quickly withdraw from the market un-throttled, failing which market participants might be tempted to add a risk premium on their orders.

Figure 1

b) **Incentive creating techniques** such as increasing the tick-size thereby making it more expensive to buy priority (price-time priority). This encourages market participants to enter orders into the book, which in turn is also a good way to increase liquidity. Another way is to encourage the use of a pricing scheme where the passive party of a trade pays less or sometimes even gets paid for participating in a trade.

c) **Prevent market makers from being discouraged** – The best (and sometimes, even extreme) way to protect market makers from alpha-creating algorithms is to simply deny them with access to relevant market segments – resulting in loss of liquidity in the marketplace.

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d) **Prevent institutional investors from being discouraged** – There are various ways to prevent information leakage when large orders originating from institutional investors are executed. One such way is the use of dark pools, which limit the transparency of large orders by not disseminating any pre-trade information, thereby lowering the market impact of these orders.

Primarily, dark pools facilitate block trading by some of the large institutional investors and hedge funds, who do not wish to impact the markets with their large or bulk orders and consequently obtain adverse prices for their trades. Lately, they have gained popularity (or negative publicity) because of their widespread creation and usage in markets. That said, it would be informative to note that there is a legitimate trade-off between protection of institutional investors executing large orders and simply putting other investor categories at a disadvantage by reducing the market transparency too much.

However, fact remains that the creation of dark pools results in a lack of transparency, which makes them vulnerable to potential conflict of interest by their owners and encourage predatory trading practices by some high-frequency traders.

5.2 Technology framework options

An important aspect to consider when accommodating algorithmic trading in markets is to evaluate the various technology options and frameworks available which, can be leveraged effectively to implement an algo-trading system or platform that broadly serves the following purposes –

a) To effectively make trading decisions
b) To submit market orders (incl. special-purpose orders) and;
c) Manage those orders after submissions

A few of those technology options include:

- *AlgoTrader* – An open source algorithmic trading platform that can be deployed in place of automated trader components.
- *Apache River* – A tool-kit used to develop distributed systems usually used as a framework for building applications.
- *Apache Hadoop* – Where pervasive logging is a requirement, then the use of Hadoop offers an interesting solution which, can be easily deployed in a clustered environment supporting CUDA technologies.
- *CUDA* – Supports high performance computational finance modelling where one can possibly achieve up to 100x performance improvements in running Monte Carlo simulations on the GPU – an open source client that allows users to share CPU-resources on a peer-to-peer based cluster – instead of the CPU.
- *FIX Engine/Protocol* – A standalone application that supports the Financial Information Exchange protocols including FIX, FAST, and FIXatdl.

5.3 Conceptual Architecture Framework

From a system or architecture perspective, there is neither a 'one-size fits all' approach nor a 'single right way' approach to conceptualizing an architecture framework for the implementation of an algo-trading system or platform.

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In fact, in HFT firms and algo-trading desks there is no consensus on the 'ideal' architecture, given that it is still an evolving discipline that is intended to meet the ever changing (dynamic) functional and non-functional requirements of systems and applications.

- **Functional requirements (FR)**, in this context refer to the expected functions of the system and its various components e.g. whether to split or splice an order, make trading decisions etc.

- **Non-functional requirements (NFR)** are measures through which the quality of a system or an application can be measured e.g. make millions of trades per second (performance), create logs for all trades made (traceability, auditability).

It would be useful to address the following aspects when eliciting and capturing the non-functional requirements in designing and developing a conceptual architecture for an algo-trading system/platform:

a) **Performance** – An algo-trading system should be able to process data both efficiently and accurately while maintain a high network throughput.

b) **Scalability** – The ability for an algo-trading system to perform (or continue performing) under expanded workloads that could arise due to increasing data feeds, number of instruments traded, number of concurrent users etc.

c) **Reliability** – Mistakes (by way of bugs) in automated trading can be costly, sometimes leading to large financial losses, fines, penalties and even suspension of trading licenses for HFT firms and algo-trading desks of Investment banks. Hence, the trading systems must be reliable.

d) **Modifiability** – An algo-trading system should be flexible enough to allow for modifications – for eg: trading strategies, rules for data pre-processing, management and financial reporting etc.

e) **Auditability** – There is increasing regulatory pressure on high-frequency trading systems to comply with new laws and regulations and therefore, they should be traceable and auditable from both a financial and IT stand-point.

f) **Security** – System and application security against cyber-threats and crimes is paramount. Trading strategies, algorithms are (usually) proprietary in nature and represent valuable intellectual property (IP). Hence, algo-trading systems should accommodate for the highest level of security, confidentiality and integrity.

g) **Fault tolerance** – This is an extension of the reliability aspect outlined above, with one major difference – the algo-trading system must continue to be reliable even after a fault or crash, in order to mitigate the risk of financial losses.

h) **Interoperability** – Algo-trading systems should be designed and architectured for easily interfacing with a wide and diverse range of related (internal and external) systems and applications e.g. order management systems, portfolio management systems, accounting applications, risk management applications and even the core-banking systems.

5.3.1 Conceptual Architecture Model

At a conceptual level, the proposed architecture (*refer Figure 2*) consists of four layers (that feed into one another) and two cross-cutting components across all the four layers.

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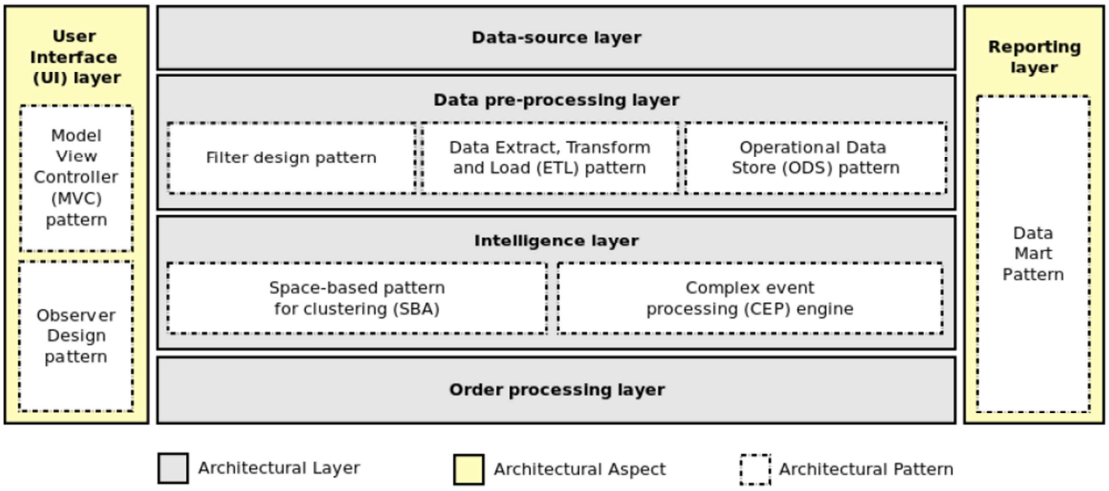

Figure 2

- **Source data layer** – This is basically the staging area where data from various external sources is received and is sent for pre-processing to the pre-processing layer. Components on this layer manage the performance of the network connections from external data sources (*Bloomberg, Reuters, D&B* etc.)

- **Pre-processing layer** – Data is received from the source data layer and is prepared for further analysis through:
    - **Filtering** – by filtering out (eliminating) irrelevant data and then passing on to the Intelligence layer as a stream of real-time events.
    - **ETL** – **extracts** filtered data, **transforms** them into a standard format/schema for it to be **loaded** to the end-target system/ODS.
    - **ODS** – an operational data store that stores relevant data and uses pre-defined business rules/logics to ensure integrity of that data only to be analyzed using a continuous query language.

- **Intelligence layer** – This is the core engine where algo-trading happens through a complex event processing engine that allows multiple event-driven trading strategies to be defined and executed in real-time.

- **Order processing layer** – Responsible for receiving trading orders from the Intelligence layer and submitting those orders to market exchanges, dark-pools, brokerages and banks. Additionally, also tracks all open positions through responses received from these entities.

- **User interface layer** and **Reporting layer** – Cuts across all the four layers because of the need to configure and manage each of those layers. Additionally, they require interfacing with **Testing** (for back-testing and stress-testing new trading rules, strategies, scenarios and simulations) and **Audit** (to generate financial and regulatory reports that confirm to IFRS and other applicable relevant standards). The reporting layer should additionally allow for 'slice and dice' and 'what-if' functionalities which, ensure compliance with both global and regional/local regulatory reporting requirements.

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It is important for an algo-trading system to satisfy both the functional and non-functional requirements in order to meet the desired user and stakeholder expectations. Some of the 'real-world' architectural constraints in designing such a system include –

- **Geographic constraints** – Physical location of the algo-trading system is becoming increasingly important – especially in the context of low-latency trading. This clearly impacts performance. Hence, many HFT firms are (already) co-locating their systems to house them inside the exchanges (BATS, NYSE, ISE etc.) which, results in some infrastructure limitations viz. cabling, sever capacity sizing, optimum hardware configuration etc.
- **Organizational constraints** – There are several organizational constraints including budgets, resources, system dependency, vendor lock-ins, program management and governance, internal policy compliance etc.
- **Regulatory constraints** – Compliance requirements vary from country to country and in the US – even from state to state. Additional regulations governing specific exchanges and securities will also constrain the algo-trading system, where applicable.

In addition to functional and non-functional requirements there are also access and integration requirements that must be considered.

- **Access requirements** define the way in which end-users interact with the algo-trading system/platform e.g. traders, system administrators, and auditors.
- **Integration requirements** specify how these algo-trading systems must interact and interface with several external systems such as Exchanges, Brokerages, Clearing and Settlements house and Banks. These interactions are mostly controlled by standard protocols such as FIX, FAST, FIXatdl, XML, and SWIFT or application programming interfaces (API's) such as the *Bloomberg Open API*, *Stocktwits*, *Thomson Reuters* etc.

5.4 Special-purpose orders

Most algo-trading systems/platforms and technology architectures support certain complex special-purpose order types and categories that are usually used in implementing both αP strategies and αC strategies. Such offerings serve two broad purposes –

a) It lowers the transaction flow thereby, reducing the cost of handling it.
b) It generates additional liquidity because certain (special purpose) order types could be implemented in a functionally superior way than standard order types. In some instances, execution risk is removed too.

5.4.1 Special-purpose order types

Some of the common and frequently implemented special-purpose order types are outlined below –

a) **Pegged orders** – This is an order in which the price is linked to some sort of external reference. So, whenever the external reference price is changed, the pegged order is automatically updated unless and until the cap-price (max. price) has been reached.

b) **Combo orders** – A combination order makes it possible for an end-user to enter a combined order like "*Buy instrument X, Sell instrument Y at the nett price P*". An illustrative order management and execution screen is shown in *Figure 3* below. From an algo-trading perspective, such orders make it easier to implement certain categories of algorithms that could result in

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effectively eliminating any arbitrage possibilities, which could prove either advantageous or disadvantageous – depending on one's view-point.

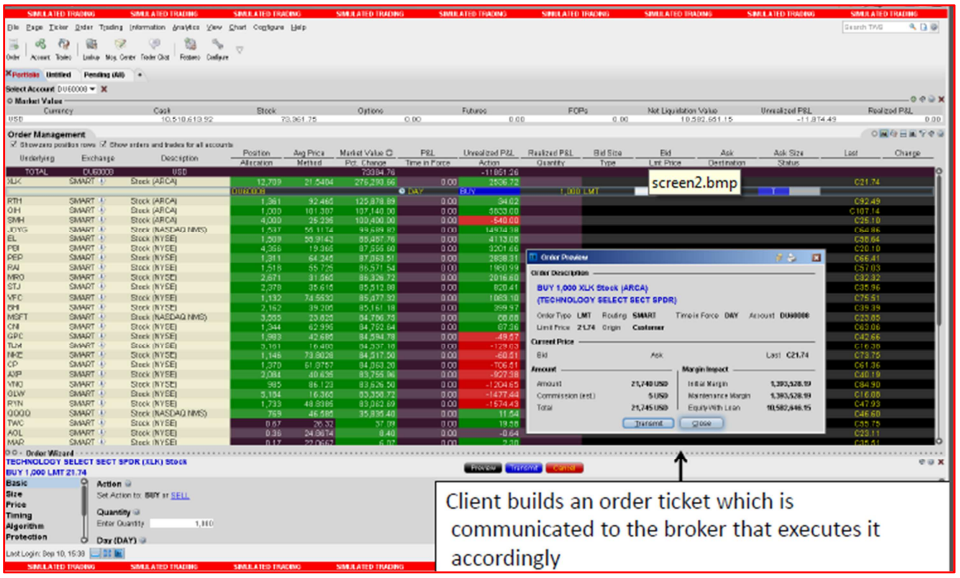

Figure 3

c) Linked orders – Basically, this is the theoretical opposite of combo orders wherein each leg has an individual price and volume, but an execution in one of the legs automatically decreases the volume in the other legs proportionately. In certain instances, a linked order is considered valuable where a trade in one of the instruments automatically reduces the quote-size in the other, thereby effectively reducing or eliminating risk.

6. CONCLUSION

Lately, algo-trading is seen to be gaining immense popularity while contributing to a significant portion of liquidity in the capital markets today. There are two primary objectives behind any algo-trading strategy that are fundamental and arise from our understanding of financial markets namely – to preserve alpha [αP] and to create alpha [αC]. Both these strategies have varying implications across markets that need to be taken into consideration as outlined in paras 5.1.1 and 5.1.4.

Being a very recent, dynamic and evolving phenomenon, capital markets are compelled – either by design or desire – to continuously monitor, analyze and report on various algo-trading strategies in order to be able to provide a well-functioning market that attracts sufficient liquidity and enables market-making. This paper attempts in outlining some of the implications of algo-trading and strategies to minimize its potentially adverse effects as highlighted in paras 5.1.2 and 5.1.5

Accommodating algo-trading in an effective manner puts a lot of demand on trading systems as mentioned in paras 5.1.3 and 5.1.6. This is pertinent in the context of certain complex special-purpose order types – illustrated in sections 5.4 and 5.4.1 – that are frequently used in implementing both αP strategies and αC strategies. It is therefore essential for market participants and operators to be provided with a high degree of flexibility and adaptability when designing a market model that is optimal and fit for use in a particular chosen market.

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Algo-trading is a technology intensive play and it's important for HFT firms and Investment banks to comprehend the various technology options available in the marketplace as highlighted in section 5.2, in order to define, conceptualize and implement a robust and scalable solution that sufficiently leverages 'best-of-breed' technologies for increased throughput and performance in the areas of automated trading.

When evaluating various HFT technology options for deployment, the fundamental principles behind architecting such a solution must neither be forgotten nor disregarded – especially during the initial requirements gathering phase – so as to comprehend and capture both the functional and non-functional requirements for development and implementation. Hence, the need to understand the associated conceptual architecture framework as illustrated in section 5.3, and the architecture model/components as depicted in section 5.3.1 (*Figure 2 on Page 10*).

From my perspective, an interesting and emerging area of research would be to analyze the implications of algo-trading on both – human and machine behavior, a topic that is very intriguing and one that has wide-ranging applications to modern societies, in the context of behavioral finance, investor heuristics and machine learning.

Certainly, there is more scope for immediate future research in the following areas and clearly, more work needs to be done in –

a) Conceptualizing algo-trading frameworks to jointly determine an optimal trading structure, given the forward looking statements/information.
b) Predicting the intra-day returns for certain asset classes and instruments (equity, debt, derivatives etc.) using certain types of algorithms and predictability models.
c) Assessing the impact of algo-trading:
    i. In *price discovery*, by way of monitoring the market for liquidity and deviations of price from fundamental value.
    ii. On *volatility* – i.e. whether the presence of algo-trading is associated with disruptive market behavior in the form of increased volatility.
d) Comprehending the economic and empirical importance of automated low-latency market activity.

In the aforementioned research areas, it's important to adequately evaluate various technology options and consider suitable architectural frameworks to realize desired outcomes to help high-frequency/algorithmic traders make more than a reasonable return on their investments.

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